Learning Support Assistant

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1. Abstract

This project focuses on developing an AI-powered learning support assistant using an agentic RAG architecture. The system classifies queries, retrieves knowledge from a vector database, and generates structured explanations. It emphasizes conceptual learning and supports follow-up queries using memory. The project demonstrates integration of LLMs, vector databases, and workflow automation tools.

2. Introduction

With the increasing adoption of AI in education, there is a growing need for systems that assist learning rather than replace thinking. Most AI tools provide direct answers, which can hinder conceptual understanding.

This project addresses that gap by building a **Learning Support Assistant** that:

- Encourages step-by-step learning
- Avoids direct exam answers
- Adapts responses based on difficulty level
- Maintains conversational context

- Technologies Used

- **n8n** – Workflow automation
- **Ollama (LLaMA 3.2)** – Local LLM inference
- **OpenAI GPT model** – Response generation
- **Supabase Vector Store** – Knowledge retrieval
- **PostgreSQL Memory** – Conversation tracking
- **Embeddings (nomic-embed-text)** – Semantic search

Training Received on Topics during first 3 weeks of training:

- Statistics for Data Science
- Data Visualization
- Introduction to Github and Cloud computing
- Streamlit (widget handling and database integration)
- Machine Learning (Regression and Classification)
- LLM fundamentals
- Communication Skills
- Deep Learning
- Computer Vision
- Agentic AI

- System view of DS

3. Project Objective

- Develop an AI-based assistant focused on conceptual learning
- Implement query classification for structured understanding
- Build a RAG pipeline for knowledge retrieval
- Integrate vector database (Supabase) for semantic search
- Enable multi-agent architecture for dynamic responses
- Maintain context using conversational memory

4. Methodology

Here is the processes our system follows to generate required response:

1. User Query Trigger

- Input is received via chat trigger node

2. Query Classification (AI Agent)

- The first AI agent classifies input into:
    - Subject
    - Topic
    - Concept
    - Difficulty
    - Intent
- Output is strictly structured JSON

3. JSON Parsing

- A custom JavaScript node:
    - Cleans malformed JSON
    - Parses structured output
    - Ensures consistency for downstream nodes

4. Intent-Based Routing

- IF node routes query based on:

    - concept_learning
    - homework_solving/exam_solving

5. Knowledge Base Creation (RAG Pipeline)

Data Pipeline:

- PDF downloaded from Google Drive
- Text split into chunks (300 size, 30 overlap)
- Converted into embeddings
- Stored in Supabase vector DB

6. Knowledge Retrieval

- Vector Store linked with AI node as tool
- Query passed as "input" to vector store
- Top-K (8) relevant chunks retrieved

7. Response Generation (Multi-Agent System)

Two agents:

Agent 1:

- Handles conceptual explanations
- Uses retrieved knowledge

Agent 2:

- Handles restricted queries (exam/homework)
- Direct refusal to direct answers
- Encourages for step-by-step concept learning

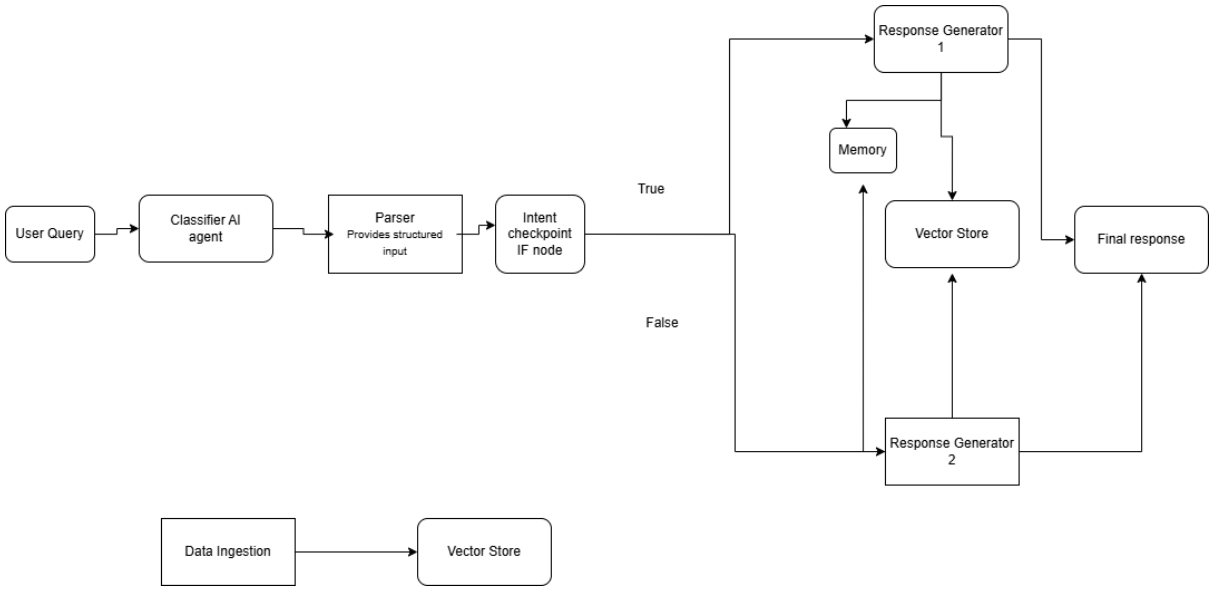
8. Memory Integration

- PostgreSQL memory stores chat history
- Enables follow-up understanding

9. Final Response

- Responses merged and returned to user

Image below is a flowchart of the implemented workflow.



5. Data Analysis and Results

- **Improved Relevance**
  Vector search significantly improved contextual accuracy

- **Better Learning Outcomes**
  Structured explanations enhanced conceptual clarity

- **Intent Handling**
  Successfully distinguished between:

  - Learning queries

  - Exam/problem-solving queries

- **Memory Efficiency**
  Follow-up questions handled effectively using context

6. Conclusion

The project successfully demonstrates the development of an AI-powered learning assistant that prioritizes understanding over answer delivery.

Key Achievements:

- Built a fully functional agentic RAG pipeline
- Integrated LLMs with vector databases
- Implemented intent-aware response system
- Enabled context-aware conversations using memory

Future Improvements:

- Add multi-document ingestion
- Improve retrieval ranking algorithms
- Introduce user feedback learning loop